NAU NORTHERN ARIZONA UNIVERSITY
Steve Sanghi College of Engineering

Memorandum

Design4Practice (D4P) Program

to: course staff

from: project team

context: undergraduate engineering project

Re: MA 1 - Practice with Variables, Data Types, and Serial IO

1.1)What is the maximum and minimum value of a character, short, and integer variable type?

The minimum and maximum values of a character are -128, and 127. For shorts and ints, the rage is -32,768 to 32,767. This can vary between boards, but the Arduino Uno we will use in this class has these limits.

1.2) What happens when you print a character data type to the serial monitor without using the DEC format?

If you print a character data type to the serial monitor without using the DEC format, the monitor will treat the output as a character code, and display the corresponding ASCII, not a number.

1.3 What happens if you declare any of these variables as unsigned?

If you declare any of these variables as unsigned, it will only be positive, or 0. It also increases the available numbers, as space does not have to be reserved for negative numbers, only 0-bit limit.

2.1) Figure 1 shows the structured outputs of the sketch. Figures 2a-d show the islolated graphs for t, f, g, and h, as Tinkercad does not have a serial plotter similar to the Arduino IDE, and this assignment was completed using a simulated Arduino Uno.

Serial Monitor

t (sec)	f(t)	g(t)	h(t)
0.00	0.00	0.00	inf
0.10	0.06	0.01	9.80
0.20	0.20	0.04	9.71
0.31	0.29	0.10	9.62
0.41	0.22	0.17	9.71
0.52	-0.06	0.27	9.52
0.62	-0.42	0.38	9.71
0.72	-0.71	0.52	9.71
0.83	-0.73	0.68	9.62
0.93	-0.40	0.86	9.71
1.03	0.22	1.07	9.62
1.14	0.86	1.29	9.71
1.24	1.24	1.54	9.62
1.34	1.12	1.81	9.71
1.45	0.47	2.09	9.71
1.55	-0.50	2.41	9.52
1.65	-1.37	2.74	9.71
1.76	-1.76	3.09	9.62
1.86	-1.42	3.47	9.71
1.97	-0.42	3.87	9.62
2.07	0.87	4.28	9.71
2.17	1.92	4.72	9.71
2.28	2.24	5.18	9.52
2.38	1.63	5.66	9.71
2.48	0.25	6.17	9.62
2.59	-1.34	6.69	9.71

Figure 1

2.2) The function millis() outputs an unsigned long data type. However, we store this as a float. Why?

We store the output of millis() because converting to seconds requires decimal numbers. If we kept millis() as an unsigned long, we would lose the fractions of seconds when converting, and lose precision in calculations.

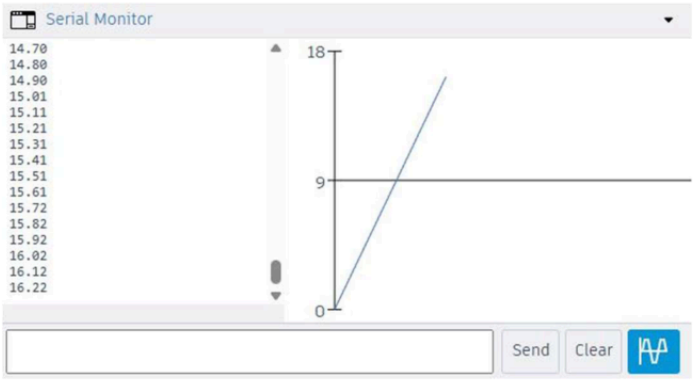


Figure 5a) t only

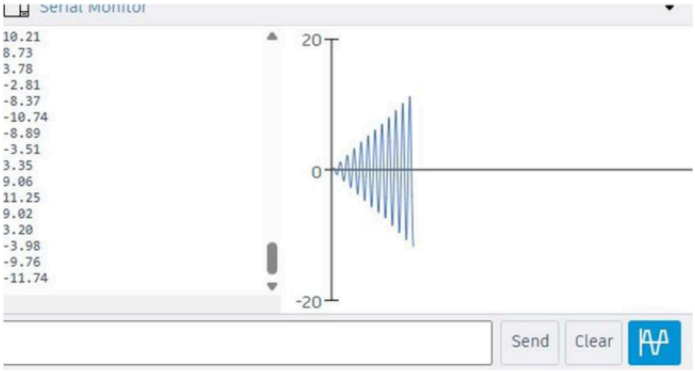


Figure 4b) f only

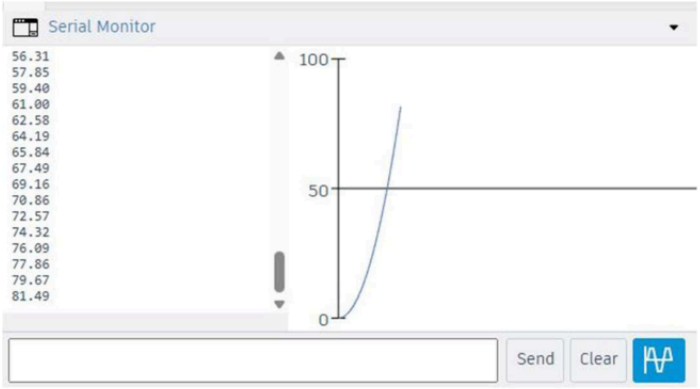


Figure 3c) g only

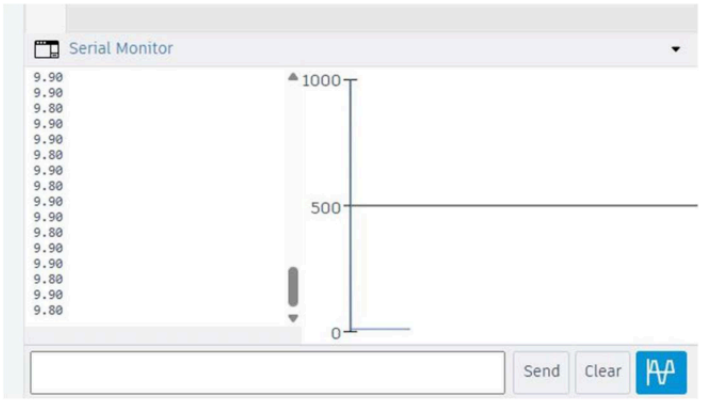
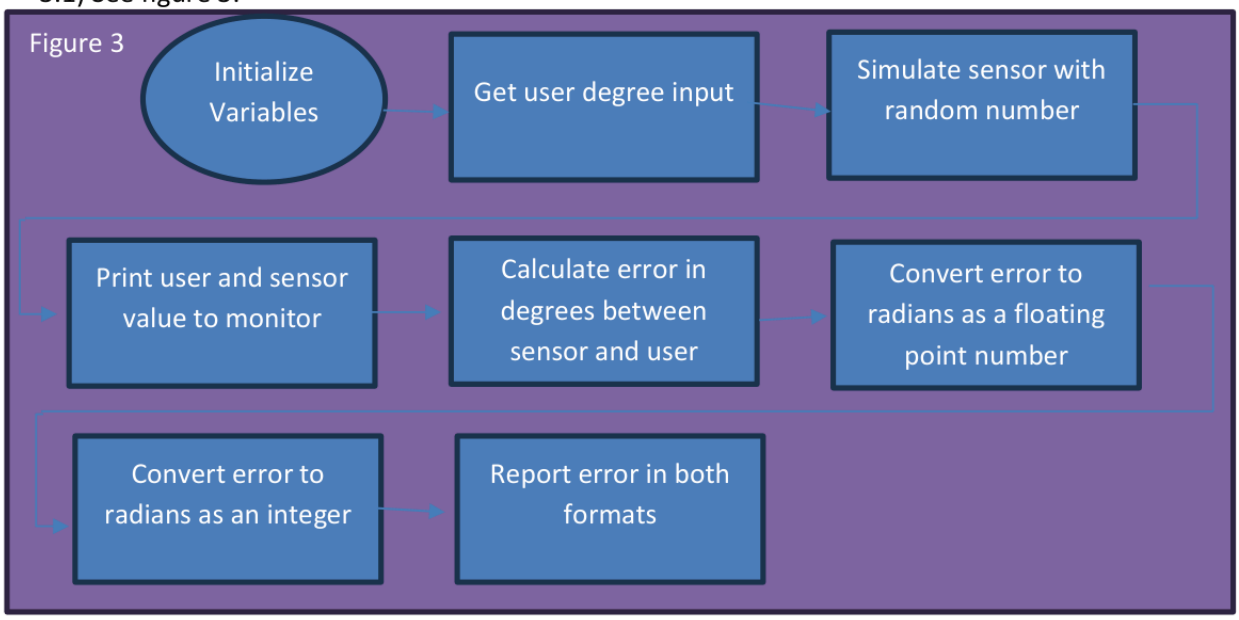


Figure 2d) h only

3.1) See figure 3.



3.2) To print the desired error in radians as both an integer and decimal, I first calculated the error in degrees by subtracting the simulated answer by the desired. Then, I initialized a new floating-point variable, errorRadFloat to convert the error to radians, by multiplying by (Pi/180). Finally, I initialized an int variable, errorRadInt, using the (int) operator on errorRadFloat.